Low-Voltage DFIG Wind Energy Trainer

Fabrication and Commissioning Booklet

Document: DFIG-LV-FAB-01 **Revision:** A **Issued to:** Fabrication contractor **Scope:** Complete supply, fabrication, assembly, wiring, commissioning and validation of one (1) low-voltage DFIG wind energy conversion trainer.

Part 0 — How to use this booklet

0.1 Structure

Work is divided into six parts, executed in order. Each part contains numbered Work Instructions (WI). Every WI has the same structure:

SAFETY — hazards, required isolation state, PPE. Read before touching anything.
Materials and tools — what must be at the bench before starting.
Procedure — numbered steps.
Acceptance check — measurable pass criteria.
Sign-off — fabricator and inspector.

Do not start a WI until the preceding WI has been signed off. The commissioning sequence in Part 6 in particular is a safety sequence, not a convenience. Skipping ahead risks equipment damage and injury.

0.2 Isolation states referenced in this booklet

State	Meaning
ISO-0	Nothing energised. Mains plug out or main isolator locked off with a personal lock. Battery disconnected at the isolator and the fuse withdrawn.
ISO-1	Control supply only (24 V DC). Mains and battery isolated and locked.
ISO-2	Battery live, mains isolated and locked.
ISO-3	Full system live. Panel doors closed. Only during commissioning steps that explicitly require it.

Every WI states its required isolation state. Lock-out/tag-out is mandatory: the person working applies their own lock and keeps the key.

0.3 Standing safety rules for the whole project

Never work alone on this rig once the battery is installed.
The 415 V primary compartment is key-locked at all times. Only the qualified electrician holds that key.
Rotating machinery guards are fitted before any shaft rotation, including hand rotation for alignment checks.
The 48 V battery can deliver over 1000 A into a short circuit. Treat every DC-side tool as a potential arc source. Use insulated tools, remove watches and rings, and never lay a tool on the battery.
A CO_2 or dry powder extinguisher rated for electrical fires is present in the work area from WI-402 onward.
Any deviation from this booklet is recorded on the deviation sheet (Appendix F) and approved by the client engineer before proceeding.

0.4 System summary (for context)

A 2 HP wound-rotor induction machine acts as the doubly-fed generator. Its stator is fed at a reduced 110 V line-to-line from an isolation transformer, so all accessible terminals stay below 120 V. Its rotor is fed through slip rings by a 48 V converter, whose DC link is a battery bank. A separate induction motor under VFD control drives the shaft to emulate variable wind speed.

Parameter	Value
Generator	2 HP, 4-pole, wound rotor, 415 V class
Synchronous speed	1500 rpm
Operating speed band	1050 – 1950 rpm
Stator supply	110 V L-L, 3-phase, 50 Hz
Stator current	3.0 A rated
Rotor standstill voltage at 110 V stator	53 V L-L
Rotor current	6.2 A rated
DC link	48 V nominal (4 × 12 V VRLA)
Shaft throughput	~400 W

PART 1 — PROCUREMENT

WI-101 — Raise purchase orders

SAFETY

No hazards. Office activity.

Procedure

Step 1. Issue orders in the sequence below. Items A and B gate the entire project and must go out on day one.

Group A — Long lead (4 to 6 weeks). Order first.

Ref	Item	Key specification	Qty	Indicative ₹
A1	Wound-rotor (slip-ring) induction machine	Per datasheet in Appendix A. **Rotor open-circuit standstill voltage 200 V L-L at 415 V stator**	1	35,000
A2	Isolation / step-down transformer	Per datasheet in Appendix B. 3 kVA, Dyn11, 110 V + 24 V secondaries	1	18,000

Group B — Import (2 to 3 weeks). Order day one.

Ref	Item	Part number	Qty	Indicative ₹
B1	Controller board	TI LAUNCHXL-F28379D	1	17,000
B2	Rotor-side inverter board	TI BOOSTXL-3PHGANINV (see note)	1	12,000
B3	Current transducer	LEM LA25-NP	3	8,000
B4	Voltage transducer	LEM LV25-P	2	5,500

Note on B2. Rotor current is 6.2 A rated and this board is rated approximately 7 A continuous. Before ordering, confirm the continuous rating with the supplier. If margin is under 25%, substitute the discrete bridge in Appendix C, which uses IRFB4110 MOSFETs and has large headroom at roughly a quarter the cost.

Group C — Domestic, stocked. Order week 2.

Ref	Item	Specification	Qty	Indicative ₹
C1	Prime mover motor	2 HP, 4-pole, 415 V, 3-ph, B3 foot mount, 1440 rpm	1	8,500
C2	Variable frequency drive	2 HP, 3-ph in / 3-ph out, sensorless vector with torque mode	1	11,000
C3	VFD brake resistor	100 Ω, 300 W, wire wound, with mounting bracket	1	1,500
C4	Incremental encoder	2048 ppr, A/B/Z, line driver, 24 V, 6 mm shaft, with flexible coupling	1	5,500
C5	Three-phase variac	0–470 V, 4 A, ganged, panel or bench mount	1	9,000
C6	Contactor	3-pole, 9 A AC-3, 230 V AC coil, 1NO+1NC aux	2	1,800
C7	MCB	4-pole, 16 A, C curve	1	1,200
C8	RCBO	4-pole, 16 A, 30 mA	1	2,800
C9	Phase failure relay	415 V, phase loss and sequence	1	1,500
C10	Emergency stop	Ø40 mm mushroom, latching, twist release, 2NC	1	800
C11	Safety relay	Dual channel, 24 V DC, forcibly guided contacts	1	4,500
C12	Power supplies	24 V 5 A DIN rail; ±15 V 1 A linear for analogue	1 each	3,500

Group D — Domestic. Order week 4.

Ref	Item	Specification	Qty	Indicative ₹
D1	Batteries	12 V, 26 Ah, VRLA SMF, with terminal boots	4	13,000
D2	DC fuse and holder	63 A Class T or BS88, DC rated 80 V minimum	1	1,800
D3	DC isolator	2-pole, 80 A, 80 V DC rated, lockable	1	2,000
D4	Pre-charge assembly	10 Ω 50 W resistor + 24 V relay	1	700
D5	Current sensor modules	ACS712-20A breakout	3	700
D6	Enclosure	800 × 600 × 250 mm, IP54, with mounting plate and lockable door	1	12,000
D7	Test terminals	4 mm shrouded banana sockets, panel mount, colour coded	40	4,000

Group E — Fabrication and consumables. Order against WI-201 once machine dimensions are confirmed. See Appendix D for the cable schedule.

Step 2. State on every order that goods are subject to incoming inspection per WI-102 and that non-conforming items will be rejected.

Step 3. For item A1, require a **routine test certificate** stating the measured rotor open-circuit standstill voltage and the equivalent circuit parameters (R1, R2′, X1, X2′, Xm) from no-load and blocked-rotor tests. These parameters are needed for controller tuning and are not optional.

Acceptance check

All orders acknowledged with confirmed delivery dates. A1 and A2 acknowledged with the full datasheet attached to the order.

Fabricator: _____ **Date:** _____ **Inspector:** _____ **Date:** _____

WI-102 — Incoming inspection

SAFETY

Isolation state: ISO-0. Received goods are not energised for inspection. Use mechanical handling aids for the machines — each weighs 30 to 45 kg. Two-person lift minimum.

Materials and tools

Digital multimeter, 500 V insulation tester (megger), vernier caliper, steel rule, camera.

Procedure

Step 1. Photograph each item as received, including packaging and any damage, before unpacking.

Step 2. Inspect the wound-rotor machine (A1):

Check	Criterion	Result
Nameplate matches order	1.5 kW, 4 pole, 415 V, 3 A	☐
Six terminals brought out	U1/V1/W1 and K/L/M on separate shrouded banana terminals	☐
Terminals engraved, not stickered	Permanent marking	☐
Slip rings and brush gear accessible	Brushes removable without dismantling	☐
Shaft runout	⩽ 0.03 mm TIR, measured on the extension	☐
Test certificate present	States rotor standstill voltage and equivalent circuit parameters	☐
Rotor standstill voltage on certificate	**200 V L-L, accept 170–230 V. Reject outside this band.**	☐
Winding insulation resistance	⩾ 5 MΩ at 500 V, each winding to frame	☐
Rotor turns by hand	Free, no rubbing or catching	☐

Reject criterion. If the rotor standstill voltage is above 230 V, the entire voltage cascade in this design is invalid and the machine must not be used without recalculation. Stop and refer to the client engineer.

Step 3. Inspect the transformer (A2): verify the nameplate shows a 110 V secondary **and** a 24 V auxiliary secondary on the same core, vector group Dyn11, and that an electrostatic shield terminal is brought out. Megger primary to secondary, primary to earth, and each secondary to earth at 500 V. All must read ⩾ 100 MΩ.

Step 4. Inspect the prime mover (C1): nameplate, free rotation, insulation resistance ⩾ 5 MΩ.

Step 5. Inspect electronic items (B1–B4, D5) visually only. Do not power. Check part numbers against the order. Store in antistatic packaging until WI-401.

Step 6. Record all measured values on the incoming inspection sheet, Appendix E.

Acceptance check

Every item on the order received, inspected, and recorded. Non-conformances raised in writing within 48 hours of receipt.

Fabricator: _____ **Date:** _____ **Inspector:** _____ **Date:** _____

WI-103 — Dimensional survey and fabrication drawings

SAFETY

Isolation state: ISO-0. Manual handling of machines. Two-person lift.

Procedure

Step 1. With both machines on the floor, measure and record:

Foot hole centres, both directions, for each machine
Foot hole diameter
Shaft centre height from the foot mounting face, each machine
Shaft extension diameter and length, each machine
Overall footprint including terminal box and fan cowl

Step 2. Note the **shaft centre height difference** between the two machines. This determines the shim pack thickness under the lower machine.

Step 3. Produce a fabrication drawing of the bed plate from these measured dimensions. Do not use catalogue dimensions. Frame sizes vary between manufacturers and a bed plate drilled to a datasheet rather than to the actual machines will require rework.

Step 4. Include on the drawing:

Slotted mounting holes, 4 mm longer than bolt diameter, in the axial direction on both machines, to allow alignment adjustment
Machined mounting pads, flat within 0.05 mm across each pad and coplanar within 0.10 mm across all four pads
An encoder mounting boss at the free shaft end of the prime mover
Four M12 tapped holes for levelling feet
An earthing boss, M8, welded to the frame

Step 5. Issue the drawing to the client engineer for approval before cutting metal.

Acceptance check

Approved fabrication drawing, dimensioned from the actual machines, signed by the client engineer.

Fabricator: _____ **Date:** _____ **Inspector:** _____ **Date:** _____

PART 2 — MECHANICAL FABRICATION

WI-201 — Fabricate the bed plate

SAFETY

Isolation state: ISO-0.

Hazards: hot work, sparks, grinding debris, fume, sharp edges, heavy sections.

Required PPE: welding helmet with correct shade, leather gauntlets, apron, safety boots, face shield when grinding, hearing protection.

Controls: hot work area screened. Fire extinguisher within 5 m. Local exhaust ventilation for welding fume. No hot work within 10 m of the battery store. Deburr all edges before the item leaves the fabrication bay.

Materials

ISMC 100 channel for the frame, 12 mm MS plate for the machine pads, 50 × 50 × 6 angle for cross members, M12 levelling feet with locknuts.

Procedure

Step 1. Cut the channel to the approved drawing. Length must allow at least 150 mm clear beyond each machine footprint for handling and guard mounting.

Step 2. Weld the frame. Fully weld all joints. Check for square: diagonals equal within 2 mm.

Step 3. Weld the 12 mm pads to the frame at the machine positions. Leave them oversize.

Step 4. Stress relieve before machining. Either thermal stress relief or a minimum 48 hour natural ageing after welding. Skipping this causes the pads to move after machining and the alignment will not hold.

Step 5. Machine the pad faces in a single setup on a milling machine so all four pads are cut coplanar. Target flatness 0.05 mm per pad, coplanarity 0.10 mm overall.

Step 6. Drill and slot the mounting holes per drawing. Slots run axially, that is, along the shaft direction.

Step 7. Weld the encoder mounting boss and the M8 earthing boss.

Step 8. Drill and tap the four M12 levelling feet positions.

Step 9. Deburr, degrease, apply one coat of red oxide primer and two coats of enamel. **Mask the machined pads and the earthing boss.** Paint on a mounting pad ruins the alignment and paint on the earth boss ruins the earth.

Acceptance check

Check	Criterion
Frame square	Diagonals within 2 mm
Pad flatness	⩽ 0.05 mm per pad, by straight edge and feeler
Pad coplanarity	⩽ 0.10 mm across all four
Hole positions	Within ±1 mm of drawing
Machined faces	Bare metal, no paint
Earthing boss	Bare metal, M8 thread clean

Fabricator: _____ **Date:** _____ **Inspector:** _____ **Date:** _____

WI-202 — Mount the machines and align

SAFETY

Isolation state: ISO-0.

Hazards: crush injury during machine placement, trapped fingers at the coupling, unexpected shaft rotation.

Required PPE: safety boots, gloves for handling, no gloves when turning the shaft by hand.

Controls: two-person lift or a chain block. Nobody places a hand between a machine and the bed plate while it is suspended. Both machines electrically disconnected and their cables not yet run.

Materials and tools

Dial indicator with magnetic base, stainless shim stock 0.05 to 1.0 mm, torque wrench, feeler

gauges, straight edge, flexible jaw coupling sized to both shafts.

Procedure

Step 1. Level the bed plate on its feet. Use a spirit level on the machined pads. Target 0.1 mm/m in both directions. Lock the levelling feet.

Step 2. Place the wound-rotor machine (A1) on its pads. Fit bolts loosely.

Step 3. Place the prime mover (C1) on its pads. Fit bolts loosely.

Step 4. Fit the coupling hubs to both shafts. Do not force them. If a hub is tight, warm it, do not hammer it — hammering drives the rotor against the bearing and brinells the races.

Step 5. Determine the shim pack. Measure the shaft centre height difference and shim the lower machine. Use the fewest shims possible; a maximum of four per foot. Shims must be full footprint, not tabs.

Step 6. Rough align by straight edge across the coupling hubs.

Step 7. Fine align with a dial indicator:

Radial (parallel) offset: mount the indicator on one hub, sweep against the other hub's outside diameter, rotate both shafts together. Target **≤ 0.05 mm TIR**.
Angular offset: sweep against the coupling face. Target **≤ 0.05 mm TIR** across the hub diameter.

Step 8. Torque the hold-down bolts progressively in a cross pattern to the values in Appendix C. Recheck alignment after torquing. Bolting almost always moves the alignment. Iterate until it holds within tolerance with the bolts fully torqued.

Step 9. Fit the coupling element.

Step 10. Rotate the assembly by hand through at least five full turns. It must turn freely and evenly with no tight spot.

Acceptance check

Check	Criterion
Radial alignment	≤ 0.05 mm TIR with bolts torqued
Angular alignment	≤ 0.05 mm TIR with bolts torqued
Hand rotation	Free, no tight spot, five turns
Bolt torque	Per Appendix C, marked with torque seal
Shim packs	≤ 4 per foot, full footprint, stainless

Fabricator: _____ **Date:** _____ **Inspector:** _____ **Date:** _____

WI-203 — Fit the encoder

SAFETY

Isolation state: ISO-0. Shaft must not rotate. Coupling guard removed for access, so a warning tag is applied to the VFD and the panel.

Procedure

Step 1. Fit the encoder mounting bracket to the boss on the bed plate at the prime mover's free shaft end.

Step 2. Fit the encoder flexible coupling to the shaft extension and the encoder shaft.

Step 3. Align the encoder body so its shaft is concentric with the machine shaft within 0.1 mm. The flexible coupling accommodates misalignment but does not eliminate the need to align.

Step 4. Set an axial gap in the coupling per the coupling manufacturer's specification. Too tight loads the encoder bearing; too loose lets it slip.

Step 5. Tighten the coupling grub screws onto the shaft flats. Apply thread lock.

Step 6. Rotate by hand. The encoder must turn with no binding and no slip.

Step 7. Route the encoder cable away from all power cables. It shares no cable tray, no conduit, and no gland plate opening with any motor or converter cable. Minimum separation 150 mm; where a crossing is unavoidable, cross at 90°.

Acceptance check

Encoder turns with the shaft with no slip through ten hand rotations. Cable routed with ≥ 150 mm separation from power cabling throughout. Marked on the cable route drawing.

Fabricator: _____ **Date:** _____ **Inspector:** _____ **Date:** _____

WI-204 — Fabricate and fit guards

SAFETY

Isolation state: ISO-0. Hot work if welding — apply WI-201 hot work controls.

Procedure

Step 1. Fabricate a coupling guard from 2 mm perforated MS sheet, perforation ⩽ 8 mm. It must fully enclose the coupling and both shaft extensions, with no opening admitting a 12 mm rod.

Step 2. Fabricate a slip-ring and brush-gear guard to the same standard. It must allow visual observation of the slip rings — this is a teaching rig and students need to see them — while preventing finger contact.

Step 3. Fabricate an encoder guard.

Step 4. Fit all guards on captive fasteners so they cannot be lost.

Step 5. Fit an interlock switch to the coupling guard, wired into the safety relay circuit in WI-303.

Step 6. Paint guards safety yellow. Apply rotating-machinery warning labels.

Acceptance check

No opening in any guard admits a 12 mm test rod. Guard interlock breaks the safety circuit when the guard is removed. Slip rings remain visible.

Fabricator: _____ **Date:** _____ **Inspector:** _____ **Date:** _____

PART 3 — ELECTRICAL PANEL FABRICATION

WI-301 — Panel layout and mechanical fit-out

SAFETY

Isolation state: ISO-0. Enclosure empty and de-energised. Drilling and punching hazards. Eye protection mandatory. Support the mounting plate when drilling; do not drill it in place inside the enclosure.

Procedure

Step 1. Divide the enclosure into four zones with physical separation:

Zone	Contents	Access
Zone 1 — 415 V	Incoming MCB, RCBO, phase failure relay, transformer primary terminals, VFD	**Key-locked sub-compartment.** Separate hinged cover, key held by the qualified electrician only.
Zone 2 — 110 V	Stator contactor, variac connections, stator test terminals	Student accessible
Zone 3 — 48 V DC and converter	DC isolator, fuse, pre-charge, converter board, DC test terminals	Student accessible
Zone 4 — Control and signal	Controller board, signal conditioning board, 24 V and ±15 V supplies, safety relay	Student accessible

Step 2. Maintain a minimum 100 mm separation between Zone 1 and all other zones. Fit an earthed metal barrier between Zone 1 and Zone 2.

Step 3. Mount DIN rail and trunking. Power and signal wiring run in **separate trunking**. They do not share a trunk at any point.

Step 4. Punch gland plate holes. Power glands and signal glands enter through **different gland plates on opposite sides** of the enclosure.

Step 5. Mount the earthing bar. This is a solid copper bar, minimum 20 × 3 mm, bolted directly to the enclosure body with a star washer through bare metal. This is the single-point star earth for the entire system.

Step 6. Mount the 4 mm test terminals on the door or on a dedicated sub-panel, grouped and colour coded:

Group	Colour	Signals
Stator voltage	Red / Yellow / Blue	U1, V1, W1, N
Rotor	Brown ×3	K, L, M
DC link	Red / Black	+48 V, 0 V
Analogue monitor	White	4 × DAC output from controller
Common	Green	Signal common

Step 7. Mount the emergency stop on the door where it is reachable from the normal operating position. Nothing may obstruct it.

Step 8. Fit the READY and FAULT indicator lamps and the RESET pushbutton.

Acceptance check

Zone 1 has an independent lockable cover. Power and signal trunking fully separate. Earthing bar bonded to bare metal. E-stop reachable and unobstructed. All test terminals labelled and shrouded.

Fabricator: _____ **Date:** _____ **Inspector:** _____ **Date:** _____

WI-302 — Power wiring

SAFETY

Isolation state: ISO-0. Mains isolator locked off with a personal lock, key retained. Battery not yet present in the panel. Verify dead with a proving unit, test the meter, test the circuit, retest the meter.

Only a qualified electrician performs Zone 1 wiring.

Materials

Cable per the schedule in Appendix D. Ferrules on every conductor end. Crimp lugs for all connections above 6 mm².

Procedure

Step 1. Wire the 415 V incomer: supply → MCB → RCBO → phase failure relay → transformer primary. All within Zone 1.

Step 2. Wire the transformer 110 V secondary → variac → stator contactor → machine stator terminals. Bring the secondary neutral to the earthing bar.

Critical. The 110 V secondary neutral is earthed at the panel star point and nowhere else. This makes the low-voltage system a separately derived, earth-referenced system with no fault path back to the 415 V supply, and it is the basis of the entire low-voltage safety case.

Step 3. Wire the transformer 24 V auxiliary secondary to a set of terminals and leave it isolated and clearly labelled **"24 V AUX — GRID SIDE CONVERTER — NOT IN USE, PHASE 6"**. Do not connect it to anything.

Step 4. Wire the 415 V VFD supply from Zone 1 and its 3-phase output to the prime mover. **Use screened VFD cable.** Terminate the screen with a 360° EMC gland at both ends.

Step 5. Mount and wire the VFD brake resistor. Locate it where its heat does not rise into the panel. Fit a guard — it reaches 200 °C in service.

Step 6. Run the rotor cable: machine slip-ring terminals → panel → converter. **Use 3-core 4 mm² screened cable.** Terminate the screen at the panel end only, to the star earth. This cable carries PWM at 20 kHz and is the single largest source of radiated interference in the rig.

Step 7. Wire the DC circuit: battery → DC isolator → Class T fuse → pre-charge assembly → DC contactor → converter. Use 16 mm² single-core with crimped and heat-shrunk ring lugs at every joint.

The fuse goes at the battery terminal end, not at the converter end. The cable itself must be protected. A fuse at the far end protects the converter but leaves several metres of unprotected 16 mm² cable connected to a source capable of over 1000 A.

Step 8. Bond every metallic enclosure, the bed plate, both machine frames, the transformer core and frame, and every cable screen to the star earthing bar with individual conductors. Do not daisy chain earths.

Step 9. Ferrule and label every conductor at both ends per the terminal schedule.

Acceptance check

Check	Criterion
Continuity of protective earth	≤ 0.1 Ω from any exposed metal part to the star bar
Insulation resistance, 415 V circuits	≥ 10 MΩ at 500 V, phase to phase and phase to earth
Insulation resistance, 110 V circuits	≥ 10 MΩ at 500 V
Phase rotation	Verified R-Y-B at transformer secondary
Every conductor ferruled and labelled	100%, both ends
Zone segregation	No power conductor in signal trunking

Do not megger any circuit connected to electronic boards. Disconnect the converter, controller and sensors first. 500 V into a microcontroller destroys it instantly.

Fabricator: _____ **Date:** _____ **Electrician (name and licence):** _____ **Date:** _____

WI-303 — Safety and control circuit

SAFETY

Isolation state: ISO-1. 24 V control supply only. Mains and battery locked off.

Procedure

Step 1. Wire the emergency stop as a dual-channel input to the safety relay. Both channels break independently.

Step 2. Wire the coupling guard interlock (WI-204) in series with the E-stop circuit.

Step 3. Wire the safety relay output contacts to drop, simultaneously:

the stator contactor
the DC contactor
the VFD run enable

Step 4. Wire the panel door interlock so opening the door drops the stator contactor.

Step 5. Wire the READY lamp to indicate: safety circuit healthy, no fault latched, DC link within range.

Step 6. Wire the FAULT lamp and latching relay. A fault latches the output off. Only the RESET pushbutton clears it. **The reset must not be automatic and must not be a maintained switch.**

Step 7. Wire the converter hardware fault output to the latching relay directly, in hardware. It also goes to the controller's trip-zone input in WI-403, but the hardware path must exist independently of any firmware.

Acceptance check

Test each of the following. Every one must drop all three outputs within 100 ms:

Test	Pass
E-stop pressed	☐
E-stop channel 1 open circuit	☐
E-stop channel 2 open circuit	☐
Coupling guard removed	☐
Panel door opened	☐
24 V supply removed	☐
Fault latches and does not self-clear	☐
RESET clears the latch only when the cause is removed	☐

Fabricator: _____ **Date:** _____ **Inspector:** _____ **Date:** _____

PART 4 — CONTROL ELECTRONICS FABRICATION

WI-401 — Build the signal conditioning board

SAFETY

Isolation state: ISO-0. ESD precautions mandatory: wrist strap bonded to the bench mat, antistatic mat, soldering iron with earthed tip. Soldering fume extraction in use. Isopropyl alcohol is flammable — no open flame.

Procedure

Step 1. Build one conditioning channel per signal. Each channel comprises:

burden resistor sized to the transducer and the measurement range
differential amplifier stage
level shift to a 0 to 3.0 V output centred on 1.5 V
second-order low-pass anti-alias filter, corner frequency 5 kHz
3.3 V clamp diodes on the output

Step 2. Set the transducer ranges:

Signal	Transducer	Jumper / turns	Full scale
Rotor current ×3	LA25-NP	Configure for 12 A primary	±15 A
Stator voltage ×2	LV25-P	Burden per datasheet	±200 V
Stator current ×3	ACS712-20A	Module as supplied	±20 A
DC link voltage	Resistive divider	20:1	0–60 V

Step 3. Fit the three LA25-NP transducers so the **rotor cable passes through the aperture**. Confirm the primary turns configuration matches the range set in step 2.

Step 4. Provide a **separate analogue ground plane** connected to digital ground at exactly one point, at the controller board's analogue ground pin.

Step 5. Power the analogue section from the ±15 V linear supply, not from the 24 V switched-mode supply. Switching noise on the analogue rail appears directly in your current measurement.

Step 6. Fit a test point on every conditioned output for oscilloscope access.

Step 7. Label every channel on the board silkscreen or with an engraved plate.

Acceptance check

Inject a known signal into each channel and verify the output:

Check	Criterion
Zero offset, all channels	1.50 V ±20 mV with zero input
Gain accuracy	Within ±2% of design, checked at 50% and 100% of range
Noise floor	⩽ 10 mV pk-pk with input shorted
Filter corner	−3 dB at 5 kHz ±20%
Clamp function	Output does not exceed 3.3 V with 150% input
Channel crosstalk	⩽ 1%

Fabricator: _____ **Date:** _____ **Inspector:** _____ **Date:** _____

WI-402 — Assemble the converter stage

SAFETY

Isolation state: ISO-0. Battery still disconnected and its fuse withdrawn.

Hazards: from this WI onward the rig contains a high-energy DC source. A dropped tool across the DC bus will arc and spray molten metal.

Required PPE: eye protection at all times in the panel. Insulated tools only on the DC side. Remove watches, rings and metal wristbands.

Controls: CO_2 extinguisher present in the work area from this point on. Battery terminals remain booted until WI-501.

Procedure

Step 1. Mount the converter board (B2, or the Appendix C discrete bridge) on standoffs, on an insulating sub-plate, in Zone 3.

Step 2. Verify heatsinking and airflow. If using the discrete bridge, fit the specified heatsink with thermal interface material and confirm the fan is wired to the 24 V supply.

Step 3. Connect the DC input with the shortest practical 16 mm² leads. **Minimise the loop area** between the positive and negative leads — run them together, twisted or closely paired, all the way to the board. A large DC loop area produces voltage overshoot at every switching transition.

Step 4. Fit the DC link capacitor bank at the converter board terminals if the board does not include adequate local capacitance. Follow the board documentation.

Step 5. Connect the three rotor phases from the panel terminals to the converter output.

Step 6. Connect the converter fault output to the hardware latch (WI-303) and to the controller trip input.

Step 7. With the battery still disconnected, verify with a multimeter that there is no short between DC+ and DC−, and no short from either DC rail to earth or to any output phase.

Acceptance check

Check	Criterion
DC+ to DC− resistance	> 1 kΩ (capacitors will charge on the meter; take the settled reading)
DC+ to earth, DC− to earth	> 10 MΩ

Check	Criterion
Each output phase to DC rails	Diode-drop readings only, symmetric across all three phases
DC loop area	Positive and negative leads paired for their full length
Fault output continuity	Verified through to the latching relay

Fabricator: _____ **Date:** _____ **Inspector:** _____ **Date:** _____

WI-403 — Install and wire the controller

SAFETY

Isolation state: ISO-1. Battery disconnected, fuse withdrawn, mains locked off. ESD precautions per WI-401.

Procedure

Step 1. Mount the controller board in Zone 4 on insulated standoffs.

Step 2. Wire the analogue inputs from the conditioning board using individually screened twisted pair. Screens to analogue ground at the controller end only.

Step 3. Wire the encoder to the eQEP input. Line driver differential pairs must remain as pairs; do not split them across a connector.

Step 4. Wire the six PWM outputs to the converter gate inputs. Keep these runs short and away from analogue wiring.

Step 5. Wire the converter fault line to the ePWM trip-zone input, not to a general purpose interrupt pin. Hardware trip acts in under a microsecond. A software interrupt does not, and the difference is a destroyed bridge.

Step 6. Wire the four DAC outputs to the white test terminals on the door, for oscilloscope observation of internal control variables.

Step 7. Wire the USB and JTAG connections out to a bulkhead connector so students can program the board without opening the panel.

Acceptance check

Continuity verified on every signal per the terminal schedule. Trip-zone path verified by shorting the fault line and observing the PWM outputs go to the safe state within one PWM period, measured on an oscilloscope.

Fabricator: _____ Date: _____ Inspector: _____ Date: _____

WI-404 — Load and bench-verify firmware

SAFETY

Isolation state: ISO-1. No power stage energised. Converter output disconnected from the rotor at the panel terminals.

Procedure

Step 1. Load the firmware. Verify the 20 kHz interrupt is running and PWM-synchronous.

Step 2. With the converter output disconnected, verify on an oscilloscope:

PWM frequency 20 kHz ±1%
Dead time correct per the converter board specification
Complementary outputs never overlap
All six channels present and correctly phased

Step 3. Inject known signals into the conditioning board inputs and verify the firmware reads the correct scaled values. Check sign convention on every channel. A reversed current sign will make the current loop positive-feedback and it will run away.

Step 4. Rotate the shaft by hand and verify the encoder count increments in the correct direction and that the index pulse is detected once per revolution.

Step 5. Verify the safe state: on trip, on reset, and on power-up, all six PWM outputs must be in the off state before any enable is asserted.

Acceptance check

Check	Criterion
Control loop period	50 µs ±0.5 µs, verified on a GPIO toggle
PWM dead time	Per board specification, verified on a scope
ADC sign convention	Correct on all nine current and voltage channels
Encoder direction	Increments for the intended forward rotation
Index pulse	One per revolution, repeatable
Safe state on power-up	All outputs off before enable

Check	Criterion
CPU load	⩽ 60% of the 50 µs budget

Fabricator: _____ **Date:** _____ **Inspector:** _____ **Date:** _____

PART 5 — INTEGRATION AND PRE-POWER CHECKS

WI-501 — Install the battery

SAFETY

Isolation state: ISO-0 becoming ISO-2 at the end of this WI.

Hazards: stored electrical energy, arc flash, acid, heavy lift. This is the highest-risk assembly step in the project.

Required PPE: face shield, insulated gloves, safety boots. Remove all metal jewellery and watches before starting.

Controls:

Insulated tools only. Every spanner used on the battery is either insulated or taped so that only the working face is exposed.
Never lay a tool on top of a battery.
Two people present.
CO_2 extinguisher within reach.
DC isolator open and locked, fuse withdrawn, throughout assembly.
Terminal boots fitted immediately after each connection is made and torqued.

Procedure

Step 1. Place the four batteries in the ventilated battery compartment. The compartment vent path directs away from the operating position.

Step 2. Verify each battery individually reads 12.6 to 13.0 V open circuit. Reject any that reads below 12.0 V.

Step 3. Connect the four batteries in series with crimped and heat-shrunk 16 mm² links. **Make the final series link last.** Torque terminals per the battery manufacturer's specification, typically 8 to 10 N·m on M8.

Step 4. Fit terminal boots on every terminal as it is completed.

Step 5. Verify the string reads 48 to 52 V open circuit, correct polarity.

Step 6. Connect the string to the DC isolator, which remains open and locked.

Step 7. Verify polarity at the converter end with the isolator open and the fuse still withdrawn.

Step 8. Insert the Class T fuse. Close and lock the compartment.

The rig is now at **ISO-2**.

Acceptance check

Check	Criterion
String voltage	48–52 V, correct polarity
Individual cell voltages	Within 0.2 V of each other
All terminals booted	100%
Terminal torque	Per manufacturer, torque-sealed
Fuse rating and type	63 A, Class T or BS88, DC rated ≥ 80 V
Compartment ventilation	Clear, directed away from operator
Isolator	Lockable, tested to open under load

Fabricator: _____ **Date:** _____ **Inspector:** _____ **Date:** _____

WI-502 — Pre-power inspection

SAFETY

Isolation state: ISO-0. Everything isolated and locked for the inspection, including the DC isolator.

Procedure

Complete the following checklist in full. **Any single failure stops the commissioning sequence.**

#	Check	Pass
1	All WI sign-offs complete to this point	☐
2	Every conductor ferruled and labelled, both ends	☐
3	No loose strands at any terminal	☐
4	All terminals torqued and torque-sealed	☐
5	Zone 1 cover fitted and key-lockable	☐
6	Barrier fitted between Zone 1 and Zone 2	☐
7	Power and signal trunking fully separated	☐
8	Protective earth continuity ≤ 0.1 Ω from every exposed metal part	☐
9	Insulation resistance ≥ 10 MΩ, electronics disconnected	☐
10	Phase rotation verified at transformer secondary	☐
11	All guards fitted, no opening admits a 12 mm rod	☐
12	Guard interlock functional	☐
13	Door interlock functional	☐
14	E-stop functional on both channels	☐
15	Battery terminals booted, fuse correct, isolator lockable	☐
16	DC cable loop area minimised at the converter	☐
17	Rotor cable screened, screen earthed at panel end only	☐
18	Encoder cable separated ≥ 150 mm from power throughout	☐
19	Coupling alignment within 0.05 mm TIR, verified after final torque	☐
20	Shaft turns freely by hand, five revolutions	☐
21	24 V AUX transformer terminals isolated and labelled	☐
22	Firmware safe state verified	☐
23	Fire extinguisher present and in date	☐

#	Check	Pass
24	Warning labels fitted: rotating machinery, electrical, battery	□

Acceptance check

All 24 items pass. Signed by the fabricator, the client engineer, and the qualified electrician. **Commissioning may not begin without all three signatures.**

Fabricator: _____ **Electrician:** _____ **Client engineer:** _____ **Date:** _____

PART 6 — TESTING AND VALIDATION

Commissioning ground rules, applying to every WI in Part 6:
Minimum two people present, one at the E-stop at all times.
Panel doors closed whenever anything is energised.
Every stage begins at the lowest setting and increases progressively.
Any unexpected noise, smell, smoke or reading means E-stop first, investigate second.
Record every measurement on the test record sheet, Appendix E. Unrecorded tests do not count as done.

WI-601 — Stage 1: Prime mover and open-circuit machine test

Purpose: verify the mechanical assembly and confirm the machine matches its certificate, before any converter is connected.

SAFETY

Isolation state: ISO-3, converter disabled.

Rotor terminals are **open circuit** at the panel. Converter DC isolator open and locked. Stator initially disconnected.

Specific hazard: the shaft rotates for the first time. Everyone stands clear of the coupling. Guards fitted.

Procedure

Step 1. Energise the VFD only. Run the prime mover uncoupled if the coupling can be easily

removed, otherwise coupled, at 10 Hz. Listen. Check for vibration.

Step 2. Ramp to 25 Hz, then 50 Hz, holding 5 minutes at each. Record bearing temperature by infrared thermometer at each step. **Stop if any bearing exceeds 70 °C.**

Step 3. Ramp to 65 Hz (1950 rpm). Confirm the VFD does not trip and vibration remains acceptable.

Step 4. Stop. Connect the stator to the 110 V supply through the variac, set to zero. Rotor still open circuit.

Step 5. Raise the variac progressively: 20 V, 40 V, 60 V, 80 V, 110 V. At each step record stator voltage, stator current and **rotor open-circuit voltage** at the slip-ring terminals, with the shaft stationary.

Step 6. Plot rotor voltage against stator voltage. Extrapolate to 110 V.

Step 7. With the stator at 110 V, run the prime mover from 1050 to 1950 rpm in 100 rpm steps. At each step record rotor voltage and rotor frequency.

Acceptance check

Check	Criterion
Bearing temperature at 65 Hz, 30 min	⩽ 70 °C
Vibration	⩽ 4.5 mm/s RMS, ISO 10816 Class I
Rotor standstill voltage at 110 V stator	**53 V L-L, accept 45–61 V**
Rotor voltage at 1050 rpm	≈ 16 V L-L
Rotor voltage at 1500 rpm	≈ 0 V
Rotor voltage at 1950 rpm	≈ 16 V L-L
Rotor frequency at 1050 and 1950 rpm	15 Hz ±1 Hz
Rotor frequency at 1500 rpm	≈ 0 Hz

If the rotor voltage at 1500 rpm does not go to approximately zero, the machine is not at synchronous speed, the pole count is wrong, or the supply frequency is not 50 Hz. Resolve before proceeding.

Result: PASS / FAIL _____ Engineer: _____ Date: _____

WI-602 — Stage 2: Converter into a dummy load

Purpose: prove the converter, firmware and protection work before the converter is ever connected to the machine.

SAFETY

Isolation state: ISO-2.

The converter output is disconnected from the rotor at the panel terminals and the disconnection is physically verified, not assumed. Stator supply isolated and locked. Prime mover isolated and locked.

Specific hazard: first energisation of the power stage. Fire risk. Extinguisher within reach. Nobody inside the panel with the DC isolator closed.

Procedure

Step 1. Connect a three-phase resistive load bank, approximately 5 Ω per phase, star connected, to the converter output.

Step 2. Verify pre-charge operates: close the DC isolator and confirm the converter DC link rises smoothly through the pre-charge resistor before the DC contactor closes.

Step 3. Enable the converter in open-loop mode at 10% modulation index, 25 Hz output. Observe load current on an oscilloscope.

Step 4. Step the modulation index: 20%, 40%, 60%, 80%. At each step record commanded and measured fundamental current.

Step 5. Verify the current sensors read correctly by comparing the LA25-NP output against a calibrated clamp meter on the same conductor.

Step 6. Test the protection. With the load connected and the converter running at 40% modulation, force a fault by asserting the fault input. Confirm the outputs go off and the fault latches.

Do not test the protection by creating an actual short circuit. Assert the fault input electrically. The purpose is to prove the trip path, not to destroy the bridge.

Step 7. Confirm RESET clears the latch and normal operation resumes.

Acceptance check

Check	Criterion
Pre-charge	DC link rises smoothly, contactor closes after, no arc
Load current waveform	Sinusoidal, THD ⩽ 5%

Check	Criterion
Modulation index linearity	Measured fundamental within 5% of commanded across 20–80%
Current sensor accuracy	Within 3% of clamp meter reading
Hardware trip	Outputs off within one PWM period
Fault latch	Does not self-clear
RESET	Clears the latch, converter restarts cleanly
Converter temperature after 15 min at 80%	⩽ 60 °C on the heatsink

The converter does not touch the machine until every line of this table passes.

Result: PASS / FAIL _____ Engineer: _____ Date: _____

WI-603 — Stage 3: Encoder alignment

Purpose: establish the rotor position reference. This is the single most common commissioning failure on a DFIG rig.

SAFETY

Isolation state: ISO-3 at reduced voltage.

Stator at 30 V L-L via the variac. Rotor open circuit. Converter disabled and its DC isolator open.

Specific hazard: the shaft is turned by hand while the stator is energised. The machine will produce torque if the rotor is not open circuit. Verify the rotor is genuinely open at the terminals before anyone touches the shaft. One person turns the shaft, a second stands at the E-stop.

Procedure

Step 1. Apply 30 V L-L to the stator via the variac. Rotor open, shaft free.

Step 2. Connect channel 1 of an oscilloscope to stator phase U1 to neutral, and channel 2 to rotor phase K to rotor neutral.

Step 3. Turn the shaft slowly by hand until the two waveforms are exactly in phase.

Step 4. Record the eQEP counter value. This is the θ_slip zero reference.

Step 5. Repeat the procedure approaching from the opposite direction of rotation. The two recorded values must agree within 10 encoder counts. If they do not, there is backlash or coupling slip — find it and fix it before proceeding.

Step 6. Write the offset to firmware non-volatile storage.

Step 7. Verify: rotate the shaft one full turn and confirm the computed slip angle behaves continuously and wraps correctly at 2π.

Acceptance check

Check	Criterion
Repeatability, both directions	Within 10 counts
Offset stored in NVM	Verified by power cycle
Slip angle continuity	No discontinuity except the intended 2π wrap

Getting this wrong swaps or inverts the d and q axes. The machine will then do something energetic and confusing on the first closed-loop attempt. Do not proceed on a marginal result.

Result: PASS / FAIL _____ Engineer: _____ Date: _____

WI-604 — Stage 4: Closed-loop rotor current, stator open circuit

Purpose: demonstrate the core DFIG behaviour with no grid connection anywhere. This stage is inherently safe because the stator is disconnected.

SAFETY

Isolation state: ISO-2 plus prime mover.

Stator disconnected at the contactor and physically verified open. The stator will develop voltage at its terminals during this test — it is an open-circuit generator. Treat the stator terminals as live. They will reach approximately 110 V.

Procedure

Step 1. Connect the converter to the rotor. Stator remains disconnected.

Step 2. Run the prime mover to 1500 rpm.

Step 3. Enable closed-loop current control with i_rd_ref and i_rq_ref both at 1.0 A. Verify the measured currents track.

Step 4. Increase progressively to 6.0 A in 1 A steps, checking converter and machine temperature at each step.

Step 5. Measure the **open-circuit stator voltage and frequency.**

Step 6. Sweep shaft speed from 1050 to 1950 rpm in 100 rpm steps, holding the rotor current constant. At each step record stator voltage, stator frequency, rotor frequency and slip.

Step 7. Apply a step change to i_rq_ref from 2 A to 4 A and capture the response on the DAC test terminals. Measure the rise time.

Acceptance check

Check	Criterion
Current tracking error, steady state	⩽ 3% of reference
Stator frequency across the full speed sweep	**50.0 Hz ±0.2 Hz at every speed**
Stator voltage stability across the sweep	Within ±5%
Current step response rise time	0.7 to 1.2 ms, consistent with 300–500 Hz bandwidth
Overshoot on step	⩽ 20%
Machine temperature after 30 min at 6 A	⩽ 80 °C on the frame
Converter heatsink after 30 min	⩽ 60 °C

This is the acceptance test for the whole project. A constant 50.0 Hz stator frequency while the shaft speed varies from 1050 to 1950 rpm is the entire DFIG concept demonstrated in one measurement. If this passes, the rig works.

Result: PASS / FAIL _____ Engineer: _____ Date: _____

WI-605 — Stage 5: Grid connection and power control

Purpose: synchronise and connect the stator, then demonstrate independent active and reactive power control.

SAFETY

Isolation state: ISO-3, full system.

Specific hazard: out-of-synchronism closure produces a large transient current and a violent torque pulse through the coupling. Verify the synchronisation check logic on the bench before arming the contactor.

Panel doors closed. Two people. One at the E-stop.

Procedure

Step 1. With the stator still disconnected, verify the PLL locks to the 110 V supply. Confirm the measured grid angle and frequency are correct and stable.

Step 2. Verify the synchronisation check logic in software: it must permit closure only when voltage magnitude is matched within 5%, frequency within 0.2 Hz, and phase within 10 electrical degrees.

Step 3. Set the variac to 40 V. Bring the machine to synchronism using rotor current control. Close the stator contactor. Record the transient current.

Step 4. If the closure is clean, open the contactor, raise the variac to 110 V, resynchronise and close again.

Step 5. With the stator connected, vary i_rq_ref and record stator active power.

Step 6. Vary i_rd_ref and record stator reactive power and power factor, holding active power constant.

Step 7. Sweep shaft speed through 1500 rpm and record the **sign of rotor power** on each side of synchronous speed.

Acceptance check

Check	Criterion
PLL lock time	⩽ 100 ms
PLL frequency error	⩽ 0.05 Hz steady state
Sync check blocks closure when out of tolerance	Verified by deliberate mismatch
Closure transient current	⩽ 1.5 × rated stator current
Independent P control	P varies with i_rq at constant Q
Independent Q control	Power factor moves lagging → unity → leading at constant P

Check	Criterion
Rotor power sign reversal	Reverses through 1500 rpm, confirming sub/super-synchronous operation
1 hour continuous run	No trip, no overtemperature, no alarm

A violent closure means the sync check is wrong. Do not repeat it hoping for a better result. Stop and fix the logic.

Result: PASS / FAIL _____ Engineer: _____ Date: _____

WI-606 — Handover

SAFETY

Isolation state: ISO-0 for the physical handover inspection.

Procedure

Step 1. Compile the handover pack:

All WI sign-off sheets
Incoming inspection records (Appendix E)
Machine and transformer test certificates
As-built wiring diagrams and terminal schedule, marked with any field changes
Firmware source, build instructions and the stored encoder offset value
All test records from WI-601 to WI-605
Deviation sheets (Appendix F) with approvals
Manufacturer manuals for the VFD, converter, controller and transducers

Step 2. Produce the operator instruction card, laminated and fixed to the panel door, covering: pre-start checks, the start sequence, normal shutdown, emergency shutdown, and fault reset.

Step 3. Conduct operator training with laboratory staff: two sessions minimum, covering safe operation, the commissioning sequence logic, and fault diagnosis.

Step 4. Demonstrate all five commissioning stages to the client engineer.

Step 5. Hand over the Zone 1 key to the nominated qualified person against signature.

Acceptance check

Handover pack complete. Training delivered and signed. Zone 1 key transferred against signature.

Fabricator: _____ Client engineer: _____ Date: _____

APPENDICES

Appendix A — Wound-rotor machine datasheet

Issue this text verbatim to the machine supplier.

Three-phase wound-rotor (slip-ring) induction machine
Output: 1.5 kW (2 HP), continuous, S1 duty
Poles: 4. Synchronous speed 1500 rpm at 50 Hz
Stator: 415 V, 3-phase, 50 Hz, star connected, rated current 3.0 A
Rotor: wound, three-phase, star connected. Open-circuit standstill voltage 200 V line-to-line, measured with rated 415 V applied to the stator, rotor open and stationary.
Rotor rated current: 6.5 A
Slip rings: three, phosphor bronze or gun metal, replaceable carbon brushes, accessible brush gear
Insulation: Class F, VPI impregnated
Enclosure: SPDP or IP23
Terminal box: **all six windings brought out** to separate 4 mm shrouded banana terminals, permanently engraved U1/V1/W1 and K/L/M
Mounting: B3 foot, single shaft extension
Rotor dynamically balanced to G2.5
Rotor resistance starter not required
Required with delivery: routine test certificate stating the measured rotor open-circuit standstill voltage, and equivalent circuit parameters R1, R2′, X1, X2′, Xm from no-load and blocked-rotor tests

Do not specify a turns ratio. Suppliers interpret ratio direction inconsistently, and a reversed interpretation produces 830 V on the slip rings instead of 200 V. Specify the measured open-circuit voltage and require the certificate.

Appendix B — Transformer datasheet

Issue this text verbatim to the transformer supplier.

Three-phase dry-type isolation and step-down transformer
Rating: 3 kVA continuous, dry type, air self-cooled, Class H insulation
Primary: 415 V, 3-phase, 50 Hz, delta

Secondary 1: 110 V line-to-line, star, neutral brought out, rated 2.5 kVA
Secondary 2 (auxiliary): 24 V line-to-line, star, **on the same core and with the same phase reference as Secondary 1**, rated 0.5 kVA
Vector group: Dyn11
Full galvanic isolation between primary and both secondaries
Electrostatic shield between primary and secondary windings, shield terminal brought out to the terminal block
Enclosure: IP24, floor mounting, shrouded barrier terminal block
Routine test certificate required: ratio, polarity, insulation resistance, no-load loss

The 24 V auxiliary secondary supports the optional grid-side converter. It costs little now and cannot be retrofitted without replacing the transformer. Order it regardless of whether the grid-side converter is planned.

Appendix C — Torque values and discrete bridge option

C.1 Fastener torque

Size	Property class 8.8	Application
M4	1.5 N·m	Board standoffs, small brackets
M5	3 N·m	Encoder bracket, guard fasteners
M6	5 N·m	DIN rail, small components
M8	12 N·m	Machine hold-down (small frames), earthing
M10	25 N·m	Machine hold-down, bed plate feet
M12	45 N·m	Levelling feet

Battery terminals: per battery manufacturer, typically 8 to 10 N·m on M8. Terminal blocks and MCBs: per component manufacturer marking. Mark every torqued fastener with torque seal.

C.2 Discrete converter bridge (alternative to B2)

If the BOOSTXL board's current rating proves inadequate:

Item	Specification	Qty
MOSFET	IRFB4110, 100 V, 180 A, TO-220	6
Gate driver	IR2110 half-bridge, or equivalent	3
Bootstrap diodes	UF4007 or similar fast recovery	3
Gate resistors	10 Ω, 1 W	6
DC link capacitors	4700 µF, 100 V, low ESR electrolytic	2
Snubber	0.1 µF, 100 V film across each leg	3
Heatsink	1 °C/W with 24 V fan	1
Current shunt	1 mΩ, 5 W, or use LA25-NP inline	3

Layout requirements: minimise the DC loop area between the capacitors and the switches; keep gate loops short and paired with their returns; provide a solid ground plane; keep gate drive and power return currents on separate paths.

Appendix D — Cable schedule

Ref	From	To	Type	Size	Screen
W01	Mains supply	Panel MCB	Armoured	4C × 2.5 mm²	Armour earthed
W02	MCB → RCBO → phase relay	Transformer primary	Panel wire	2.5 mm²	—
W03	Transformer 110 V sec	Variac	Panel wire	2.5 mm²	—
W04	Variac	Stator contactor	Panel wire	2.5 mm²	—
W05	Stator contactor	Machine stator	Flexible	4C × 2.5 mm²	—
W06	Machine slip rings	Panel rotor terminals	Flexible	**3C × 4 mm² screened**	Panel end only

Ref	From	To	Type	Size	Screen
W07	Panel rotor terminals	Converter output	Panel wire	4 mm²	Twisted
W08	Battery	DC isolator	Single core	16 mm²	—
W09	DC isolator → fuse → contactor	Converter DC in	Single core	**16 mm², paired, minimum loop area**	—
W10	Zone 1	VFD input	Panel wire	2.5 mm²	—
W11	VFD output	Prime mover	**VFD screened cable**	4C × 2.5 mm²	360° EMC gland both ends
W12	VFD	Brake resistor	Heat resistant	2.5 mm², silicone	—
W13	Encoder	Controller eQEP	**Shielded twisted pair**	6C × 0.5 mm²	Controller end only
W14	Transducers	Conditioning board	**Individually screened pair**	0.5 mm²	Analogue ground, one end
W15	Conditioning board	Controller ADC	Screened pair	0.5 mm²	Analogue ground, one end
W16	Controller PWM	Converter gate inputs	Ribbon or twisted	0.5 mm²	Short as possible
W17	Safety circuit	All devices	Panel wire	1.0 mm²	—
W18	All frames and screens	Star earthing bar	Green/yellow	4 mm² individual	Never daisy chained

Appendix E — Test record sheets

Reproduce one copy of each for the project file.

E.1 Incoming inspection record — item, order ref, date received, condition, measured values, pass/fail, inspector, signature.

E.2 Alignment record — radial TIR before and after torque, angular TIR before and after torque, shim thickness per foot, ambient temperature, engineer, signature.

E.3 Insulation and earth continuity record — circuit, test voltage, measured resistance, pass/fail, instrument serial number, calibration date, electrician, signature.

E.4 Stage 1 record — speed, stator voltage, stator current, rotor voltage, rotor frequency, bearing temperature, vibration. One row per test point.

E.5 Stage 2 record — modulation index, commanded current, measured current, THD, heatsink temperature. One row per test point.

E.6 Stage 3 record — encoder offset forward, encoder offset reverse, difference in counts, stored value.

E.7 Stage 4 record — shaft speed, i_rd_ref, i_rq_ref, i_rd measured, i_rq measured, stator voltage, stator frequency, rotor frequency, slip, temperatures. One row per test point.

E.8 Stage 5 record — closure transient current, i_rd_ref, i_rq_ref, stator P, stator Q, power factor, rotor power sign, shaft speed. One row per test point.

Appendix F — Deviation sheet

Field	Entry
Deviation number	
WI reference	
As specified	
As proposed	
Reason	
Effect on safety	
Effect on performance	
Raised by / date	
Client engineer approval / date	

No deviation is implemented before client engineer approval is recorded here. Deviations affecting the voltage cascade, the protective earthing arrangement, the safety circuit, or the

battery protection require written approval and a re-check of WI-502.

End of document DFIG-LV-FAB-01 Rev A.